

Curriculum Vitae

Anargha Mondal

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EDUCATION

Indian Institute of Science Education and Research Pune
Bachelor of Science-Master of Science (BS-MS), **Physics Major**

Expected Graduation: August 2027

CGPA: 8.6/10

RESEARCH EXPERIENCE

Master's Thesis Student, IISER Pune

Maharashtra, India

May 2026 - Present

- Supervised by Dr. Arka Banerjee and in collaboration with Dr. Adrian Bayer.
- Investigating whether the scale dependence of the halo bias in massive neutrino cosmologies can be detected with weak lensing observables. The goal is to then constrain the mass of neutrinos from the scale-dependent bias signal.

Research Project Student, IISER Pune

Maharashtra, India

August 2024 - Present

- Supervised by Dr. Arka Banerjee.
- Obtained stronger constraints on cosmological parameters from halos cross-correlated with the total matter field using Nearest Neighbour distributions, compared to the two-point correlation function.
- Results indicate that using the most massive halos in the investigation significantly improves constraints when cross-correlated with the total matter field, compared to using simulation particles.
- Obtained Fisher forecasts for neutrino mass, useful for upcoming large-scale structure surveys like CMB S4, Euclid, etc.

Research Intern, NAOJ, Mitaka

Tokyo, Japan

May 2024 - February 2025

- Supervised by Dr. Maria Giovanna Dainotti and Dr. Biagio De Simone.
- Developed a new joint catalogue of Type Ia supernovae by combining the best samples available: Pantheon Plus, Pantheon, DES, JLA and SH0ES.
- Performed a Markov Chain Monte Carlo analysis of the binned 'master' sample to determine the H_0 value for each redshift bin, for various Dark Energy models.
- Results indicate a decreasing trend in the Hubble Constant with redshift, which might reconcile the Hubble Tension.
- Published in the *Journal of High Energy Astrophysics (JHEAP)*.

Research Intern, TIFR Mumbai

Online

July 2023 - December 2023

- Supervised by Dr. Sudip Bhattacharyya and Dr. Yash Bhargava.
- Learned about the high-energy processes in accretion disks around black holes.
- Tested how well our current theoretical models of accretion disks in Black Hole Binaries and Low Mass X-Ray Binaries fit observed data, using data from the Large X-ray Photon Counter and Soft X-ray Telescope on the AstroSat.

SOFTWARE DEVELOPMENT

kNNpy

May 2025 - Present

- A Python package for higher-order cosmological clustering using nearest-neighbour distributions. Links to the website and the GitHub repository
- Contributed towards building the codes, ideation and design.

COMPUTATIONAL SKILLS

Languages: Python, Bash

Tools: Mathematica, MATLAB, HEASoft, Git and GitHub, L^AT_EX

PUBLICATIONS

- *A New Master Supernovae Ia sample and the investigation of the H_0 tension*
Dainotti et al, including Mondal, A. (2025). (JHEAP)

AWARDS & FELLOWSHIPS

- National Top 1% in National Graduate Physics Examination (NGPE) *February, 2023*
- Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow *August, 2021*
- National Talent Search Examination (NTSE) Scholar *March, 2021*
- Indian Olympiad Qualifier in Mathematics (IOQM) Merit Awardee *January, 2021*
- Jagadish Bose National Talent Search Scholar *December, 2020*

TALKS & CONFERENCES

Talks

- “Investigating the Hubble Tension with Type Ia Supernovae” *13th November, 2024*
Invited Talk, Science Club, IISER Pune, India.
- “Investigating the Hubble Tension with Type Ia Supernovae” *14th November, 2024*
Contributed Talk, Lagrangians to Lasers, IISER Pune, India.

Conferences

- **Pune-Mumbai Cosmology and Astro-Particle Meeting (PMCAP)** *12th – 14th September, 2025*
Sixth edition organised at NCRA, Pune, Maharashtra, India
- **Pune-Mumbai Cosmology and Astro-Particle Meeting (PMCAP)** *12th – 14th September, 2025*
Fifth edition organised at IISER Pune, Maharashtra, India
- **Pune-Mumbai Cosmology and Astro-Particle Meeting (PMCAP)** *7th – 8th February, 2025*
Fourth edition organised at IIT Bombay, Maharashtra, India
- **Conference on Blazars and Restless AGN (COBRA)** *22nd July, 2024*
Organised by IUCAA, Pune and Presidency University, Kolkata

CO-SCHOLASTIC ACTIVITIES

- Coordinator, Helicase, the Science Magazine of IISER Pune (March 2023 - February 2024)
- Physics Question-Making Team Member, Mimamsa (October 2022 - Present)
- ‘Cosmic Forums’ Team Member, Aakashganga, Astronomy Club of IISER Pune
- Regular speaker in the Lagrangians to Lasers and Cosmology Journal Clubs
- Core Team Member, Math Club, IISER Pune